A Pre-Registered Synthetic Benchmark for Open-Set Radio Signal Classification under Transmitter-Level Splits

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***Abstract*—Open-set recognition asks a classifier to label known classes and to reject inputs from classes it never saw. We present a small synthetic benchmark for open-set radio signal classification in which every nuisance factor is explicit: each simulated transmitter has its own IQ imbalance, amplifier compression, phase noise and DC offset, and data are split by transmitter. Hypotheses and pass criteria were fixed before the runs and then tested again on five fresh seeds. On four known modulations and two held-out ones, five post-hoc scores were compared on the same trained network. Euclidean distance to class centroids in embedding space reaches an AUROC of 0.779 ± 0.020 over ten seeds; maximum softmax probability, max-logit and energy reach 0.353 ± 0.088, 0.278 ± 0.065 and 0.286 ± 0.068, below chance. Mahalanobis scoring is worse than the Euclidean score in all ten seeds. Training against auxiliary outliers helps only when the outliers resemble the unknowns: with the training seed held fixed, rejection of 4PAM unknowns is 0.423 ± 0.189 with 8PAM outliers and 0.009 ± 0.014 with 16PSK outliers. Four of eight initial hypotheses replicated, two negative results replicated, one effect did not replicate, one was inconsistent between the two seed sets, and one of our own interpretations was refuted by a follow-up experiment. The benchmark is a measurement instrument on synthetic data, not a new method.**

***Index Terms*—Open-set recognition, radio signal classification, modulation classification, hardware impairments, out-of-distribution detection, pre-registration, reproducibility.**

I. Introduction

DEEP classifiers for radio signals are usually evaluated on test classes that were all present in training. A deployed monitor does not have that guarantee: new emitters, waveforms and interference types appear, and the useful answer for them is "unknown", not the nearest known label. Open-set recognition (OSR) addresses this case [1], [2].

OSR results on radio data are hard to interpret for two reasons. Public datasets entangle channel, hardware and capture session, so it is unclear what a rejection score reacts to. And hardware impairments are themselves learnable signatures, which is the basis of RF fingerprinting [3]; a split by frame therefore lets a model succeed by recognising devices.

This paper goes the other way from realism. It studies a deliberately small synthetic benchmark in which each factor is controlled and published. Each transmitter draws its impairment parameters once; data are split by transmitter; unknown classes take no part in training, model selection or threshold calibration; and the threshold is set on known validation data at a fixed acceptance target. Inside this instrument we ask how standard open-set scores behave when the unknown classes are structurally related to the known ones (64QAM to 16QAM, 4PAM to BPSK).

The second concern of the paper is the reliability of its own conclusions. Every hypothesis and its pass criterion was written down before the corresponding runs. The first eight hypotheses were then tested again on five fresh seeds, and three follow-up hypotheses addressed weaknesses that an internal review found in the first analysis. We report what replicated, what did not, and where our interpretation was wrong.

The contributions are:

1) an open benchmark for open-set radio signal classification with explicit per-transmitter impairments and transmitter-level roles for training, model selection, calibration and test;
2) a comparison of five post-hoc scores on the same trained network per seed, and of the Objectosphere loss [4] under three auxiliary-outlier pools with the training seed held fixed;
3) a pre-registered analysis with a replication on fresh seeds, including two replicated negative results, one effect that did not replicate, and one refuted interpretation.

Code, pre-registrations, per-seed results and the scripts that generate every number and figure of this paper accompany the submission.

II. Related Work

OSR was formalised for visual classifiers by Scheirer *et al.* [1] and extended to deep networks by OpenMax [5]; Geng *et al.* survey the field [2]. The scores compared here are standard: the maximum softmax probability (MSP) baseline [6], the max-logit score [7], the energy score [8], and class-conditional distances in feature space, of which the Mahalanobis detector [9] is the best-known form. The Objectosphere loss [4] trains against auxiliary known outliers. Detection quality is summarised by the area under the ROC curve [10].

Deep modulation classification was established on synthetic and over-the-air data by O'Shea *et al.* [11], [12], who showed that hardware and channel effects change the results. Open-set formulations have been applied to transmitter authorisation [13] and to modulation recognition with signal-domain features [14]. RF fingerprinting shows that impairments such as IQ imbalance and DC offset identify a device [3]; this is why the benchmark splits by transmitter. We do not propose a method and claim no superiority of any score beyond the conditions measured.

III. System Model and Problem Formulation

A. Signal model

A frame is a complex baseband sequence of $N = 512$ samples at 8 samples per symbol. Symbols are drawn uniformly from the constellation of the transmitter's modulation and shaped by a unit-energy root-raised-cosine filter (roll-off 0.35, span 6 symbols); the frame is a window taken after the filter transient. The transmitter then applies, in this order,

$$y_1[n] = x[n]\left(1 + (|x[n]|/A_{\text{sat}})^{2p}\right)^{-1/(2p)}, \tag{1}$$

$$y_2[n] = \mu\, y_1[n] + \nu\, y_1^*[n], \tag{2}$$

$$y_3[n] = y_2[n]\, e^{j\theta[n]} + d, \tag{3}$$

where (1) is the Rapp amplifier model [15], (2) is IQ imbalance with $\mu = (1 + ge^{-j\phi})/2$ and $\nu = (1 - ge^{j\phi})/2$, $\theta[n]$ is a Gaussian random walk (phase noise) and d is a DC offset. The frame then receives a carrier-frequency offset $f \sim \mathcal{U}(-0.01, 0.01)$ cycles per sample with a random initial phase, and complex white Gaussian noise at a stated signal-to-noise ratio (SNR), defined on the impaired, frequency-shifted signal.

A transmitter fixes its modulation and impairment parameters once: $g \sim \mathcal{U}(0.9, 1.1)$, $\phi \sim \mathcal{U}(-5°, 5°)$, real and imaginary parts of d from $\mathcal{U}(-0.05, 0.05)$, phase-noise standard deviation log-uniform in $[2 \times 10^{-3}, 10^{-2}]$ rad per sample, $A_{\text{sat}} \sim \mathcal{U}(1.0, 1.6)$ and $p \sim \mathcal{U}(2, 3)$. An *ideal* device sets all of these to their identity values. No fading, multipath or receiver impairment is simulated.

B. Classes and roles

The known classes are BPSK, QPSK, 8PSK and 16QAM. The unknown classes, used only at test time, are 64QAM and 4PAM. Two further classes, 16PSK and 8PAM, serve as auxiliary training outliers for the Objectosphere variants. All frames of a device stay in one role: training, validation for model selection, validation for calibration, or test. Unknown-class frames influence no fit, no model selection and no threshold.

C. Decision rule

A score $s(\mathbf{x})$, larger meaning more unknown-like, is compared with a threshold τ: the frame is accepted if $s(\mathbf{x}) \le \tau$ and then receives the label of the largest logit. τ is the smallest value that accepts at least 95% of the known calibration frames.

IV. Scores and Training Variants

The embedding network (CNN1D) has three blocks of convolution (kernel 7), batch normalisation, ReLU and max pooling with 32, 64 and 128 channels, followed by global average pooling and a linear layer; the embedding dimension is 128. With logits $\mathbf{z}$, embedding $\mathbf{e}$ and class centroids $\mathbf{m}_c$ computed from the known training frames, the post-hoc scores are

$$s_{\text{MSP}} = 1 - \max_c \operatorname{softmax}(\mathbf{z})_c, \tag{4}$$

$$s_{\text{logit}} = -\max_c z_c, \qquad s_{\text{energy}} = -\log \textstyle\sum_c e^{z_c}, \tag{5}$$

$$s_{\text{Euc}} = \min_c \|\mathbf{e} - \mathbf{m}_c\|_2, \tag{6}$$

$$s_{\text{Mah}} = \min_c \big((\mathbf{e} - \mathbf{m}_c)^{\mathsf{T}} \boldsymbol{\Sigma}_\alpha^{-1} (\mathbf{e} - \mathbf{m}_c)\big)^{1/2}, \tag{7}$$

with $\boldsymbol{\Sigma}_\alpha = (1-\alpha)\boldsymbol{\Sigma} + \alpha\,(\operatorname{tr}\boldsymbol{\Sigma}/128)\,\mathbf{I}$, where $\boldsymbol{\Sigma}$ is the pooled within-class covariance of the known training embeddings and α is the shrinkage; $\alpha = 0.1$ unless stated. At $\alpha = 1$ the Mahalanobis score ranks frames exactly as the Euclidean score does.

The Objectosphere variants [4] add outlier frames to the training set. The loss pushes the logit norm of known frames above a margin $\xi = 10$ and drives outlier frames toward uniform, small-norm logits with weight $\lambda = 0.1$; the score is s_{MSP}. Three outlier pools are compared: both auxiliary classes (800 frames), 16PSK only and 8PAM only (400 frames each).

V. Experimental Setup

A. Data, training and metrics

Per seed and known class there are 40 training, 10 validation and 10 test devices, with 20, 10 and 10 frames per device; the validation devices are halved by device into model selection and calibration. There are 20 devices per auxiliary class and 10 per unknown class. This gives 3200 known training frames, 200 calibration frames, and 400 known and 200 unknown test frames per seed. SNRs are drawn from -4 to 20 dB in steps of 2 dB. Networks are trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}, batch size 64) for at most 25 epochs, with early stopping (patience 5) on the accuracy over the model-selection devices.

The AUROC takes the unknown class as positive [10]. The unknown rejection rate is the fraction of unknown test frames with $s > \tau$; the known acceptance rate is the fraction of known test frames with $s \le \tau$. The open-set classification rate (OSCR) curve [4] plots the fraction of known test frames that are accepted and correctly labelled against the fraction of unknown frames accepted, as τ varies; we report its area. Frames of one device are correlated, so the seed is the unit of replication: all dispersions are sample standard deviations over seeds.

Two companion tasks validate the simulator. Closed-set modulation classification is run with impaired and with ideal devices, for the CNN1D and for a multilayer perceptron (MLP; hidden widths 256 and 128) on the raw frame, and tested at -6 to 20 dB. A fingerprinting task asks a one-dimensional ResNet [16] to identify 10 devices that all send QPSK; here the device is the label, so the split is by frame (600 test frames per seed), and an ideal-device control must stay at the chance level of 0.10.

B. Pre-registration and replication

The study has three campaigns. Campaign 1 trains each open-set variant separately and runs the companion tasks (H1–H4). Campaign 2 trains one cross-entropy network per seed,

evaluates the five post-hoc scores on it, and trains the three Objectosphere pools (H5–H8). Campaign 3 retrains the three pools with one shared training seed, so that only the pool differs (H7s); scores the campaign-1 classifiers on new frames of their training devices (H9, H10); and sweeps the shrinkage α, which is exploratory.

Campaigns 1 and 2 first ran on seeds 0–4 and were then repeated with unchanged code on fresh seeds 5–9. Each hypothesis predicts the sign of a per-seed paired difference. It is *supported* on a set of five seeds if the predicted sign occurs in at least four, and on ten seeds if it occurs in at least eight. We call a hypothesis *replicated* only if the fresh seeds meet the four-of-five criterion on their own. With five or ten seeds we report counts, not significance; for reference, the two-sided sign-test p-value is 0.002 at 10/10, 0.021 at 9/10, 0.109 at 8/10 and 0.344 at 7/10.

The analysis plans of campaigns 2 and 3 and of the replication were committed to the repository before their runs started. The plan of campaign 1 was written before its run but committed about 3.5 minutes after the run began; its criteria were not changed afterwards, and its hypotheses are among those retested on fresh seeds. All runs used PyTorch on four CPU threads; the five campaign runs took 1045, 749, 1093, 505 and 1176 s. No run failed and no seed was excluded.

VI. Results

Unless stated otherwise, values are means and sample standard deviations over seeds 0–9. Table II lists every hypothesis with its counts.

A. Post-hoc scores on one network (H1, H5, H6)

Table I and Fig. 1 give the main result. On the same network, the three scores computed from the logits have a mean AUROC below 0.5: 0.353 ± 0.088 for MSP, 0.278 ± 0.065 for max-logit and 0.286 ± 0.068 for energy. Max-logit and energy are below 0.5 in every seed (maxima 0.368 and 0.386); MSP is below 0.5 in nine seeds and reaches 0.509 in one. A value below 0.5 means that the unknown frames receive, on average, *more* confident outputs than the known ones. At the calibrated threshold MSP rejects a fraction 0.028 ± 0.015 of the unknown frames.

The two scores computed from the embedding separate the classes: Euclidean distance reaches 0.779 ± 0.020 and rejects 0.512 ± 0.034 of the unknown frames at a known acceptance of 0.954 ± 0.016. H1, tested in campaign 1 with separately trained networks, is supported in 5/5 original and 5/5 fresh seeds (mean AUROC difference $+0.433$); campaign 2 shows the same gap on a single network, so it is not an effect of training variance. Fig. 2 shows the mechanism for one seed: in MSP the unknown frames concentrate at the confident end, on the accept side of the threshold, whereas in distance a large part of them lies beyond it. Fig. 3 shows the corresponding OSCR curves.

Two predictions failed, and both failures replicated. H5 predicted that the Mahalanobis score would beat the Euclidean score. It was lower in all ten seeds (0/10 in the predicted direction; mean difference -0.031). H6 predicted that the energy score would beat MSP. It did so in 2/5 original and 0/5 fresh seeds (mean difference -0.067).

B. Auxiliary outliers (H2, H7, H7s)

Training against both auxiliary classes raises unknown rejection above the MSP baseline in every seed (H2, 10/10; 0.155 ± 0.057 against 0.028 ± 0.015 in campaign 1), although it stays far below the Euclidean score. The outlier class 8PAM is close to the unknown class 4PAM, which we recorded as a confound before the first run. Campaign 2 measured it by changing the outlier pool: rejection of 4PAM frames is 0.348 ± 0.225 with 8PAM outliers and 0.004 ± 0.005 with 16PSK outliers (H7, 5/5 original and 5/5 fresh seeds).

In campaign 2 the three pools were also trained with different seeds. Campaign 3 removed that difference. With one training seed for all pools, 4PAM rejection is 0.423 ± 0.189 with 8PAM outliers and 0.009 ± 0.014 with 16PSK outliers (H7s, 10/10; per-seed difference from $+0.050$ to $+0.670$). Rejection of 64QAM frames stays low under every pool (0.058 ± 0.045, 0.037 ± 0.026 and 0.088 ± 0.054; Fig. 4). The gain from outlier exposure in this benchmark is therefore transfer from 8PAM to 4PAM. It is also unstable: with 8PAM outliers the per-seed 4PAM rejection ranges from 0.050 to 0.680. The pool with 16PSK alone gives an AUROC of 0.273 ± 0.073, below chance like the plain MSP baseline.

C. Dependence on SNR (H8)

H8 predicted that the MSP AUROC would be higher at low SNR ($\leq$ 2 dB) than at high SNR ($\geq$ 10 dB), as an explanation by confidence suggests. It was supported on the original seeds (4/5) and *not replicated* on the fresh ones (3/5; pooled 7/10, below the criterion). Over ten seeds the two bands give 0.366 ± 0.082 and 0.338 ± 0.097. We conclude that the benchmark does not show an SNR dependence of the MSP failure. Fig. 5 shows the AUROC per SNR; the cells are small (at most a few tens of frames per SNR and seed), and the Euclidean score stays above the MSP score at every SNR in the mean.

D. Validation of the simulator (H3, H4)

With impaired devices the fingerprinting network identifies the device with accuracy 0.221 ± 0.052; with ideal devices it scores 0.099 ± 0.011, at the chance level of 0.10 (H3, 10/10). The impairment chain therefore carries the device identity and nothing else in the pipeline does.

H4 predicted that removing the impairments would raise the closed-set accuracy of the CNN1D. The accuracies are 0.770 ± 0.019 for ideal and 0.749 ± 0.032 for impaired devices. The predicted sign occurred in 3/5 original and 4/5 fresh seeds, 7/10 in all, which is below the ten-seed criterion; the mean difference is $+0.022$. If the impairments cost closed-set accuracy at these magnitudes, the cost is about two points and not consistently detectable with ten seeds. Fig. 6 shows the accuracy against SNR.

TABLE I

OPEN-SET RESULTS OVER TEN SEEDS (MEAN ± STANDARD DEVIATION). THE FIVE POST-HOC SCORES USE THE SAME CROSS-ENTROPY NETWORK PER SEED; THE OBJECTOSPHERE ROWS ARE SEPARATELY TRAINED NETWORKS SCORED WITH MSP. 400 KNOWN AND 200 UNKNOWN TEST FRAMES PER SEED; CALIBRATION TARGET 0.95.

Variant	AUROC	Unknown rejection	Known acceptance	OSCR area
MSP	0.353 ± 0.088	0.028 ± 0.015	0.951 ± 0.029	0.333 ± 0.083
Max-logit	0.278 ± 0.065	0.033 ± 0.019	0.944 ± 0.028	0.249 ± 0.055
Energy	0.286 ± 0.068	0.034 ± 0.014	0.943 ± 0.026	0.252 ± 0.057
Euclidean distance	0.779 ± 0.020	0.512 ± 0.034	0.954 ± 0.016	0.608 ± 0.060
Mahalanobis ($\alpha = 0.1$)	0.748 ± 0.024	0.472 ± 0.017	0.947 ± 0.022	0.587 ± 0.060
Objectosphere, 16PSK + 8PAM outliers	0.536 ± 0.061	0.194 ± 0.067	0.945 ± 0.024	0.464 ± 0.059
Objectosphere, 16PSK outliers	0.312 ± 0.070	0.033 ± 0.024	0.941 ± 0.025	0.289 ± 0.059
Objectosphere, 8PAM outliers	0.581 ± 0.068	0.206 ± 0.129	0.936 ± 0.033	0.487 ± 0.044

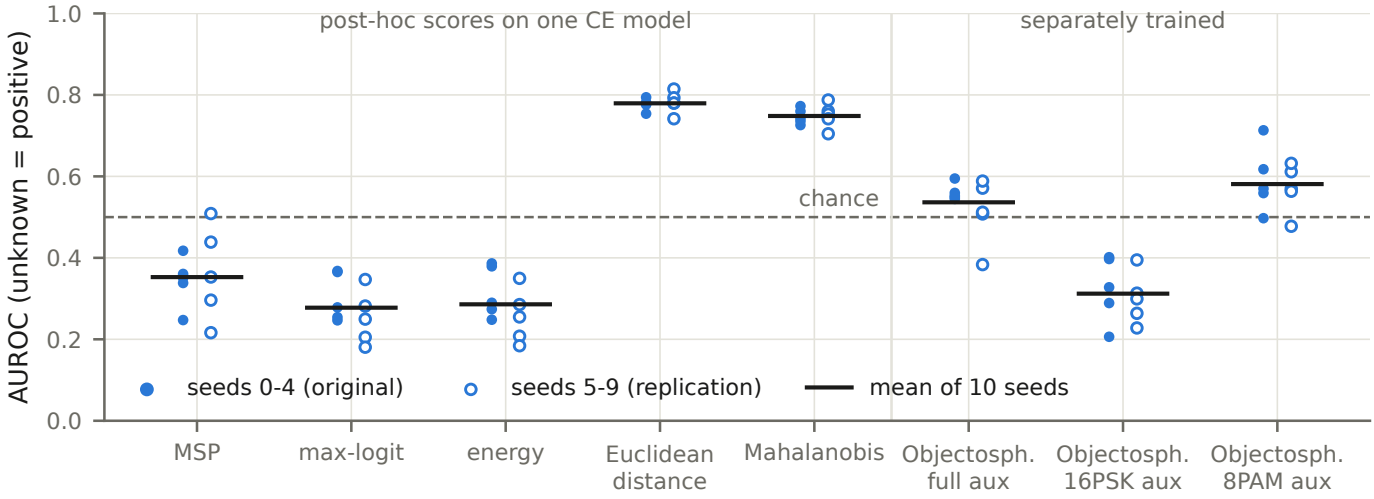


Fig. 1. AUROC of every open-set variant for each seed. Filled markers are the original seeds 0–4, open markers the replication seeds 5–9, and the bar is the mean of all ten.

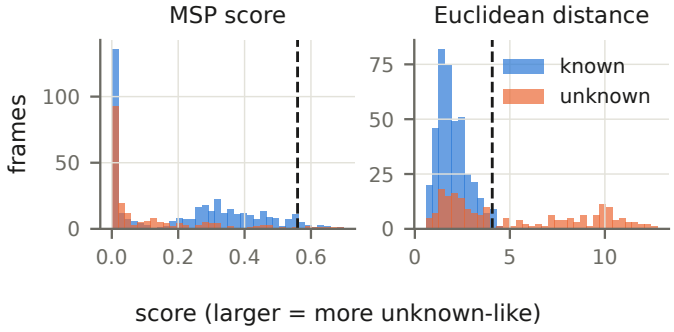


Fig. 2. Scores of the known and unknown test frames on one network (seed 0, 400 known and 200 unknown frames). The dashed line is the calibrated threshold; frames to its left are accepted.

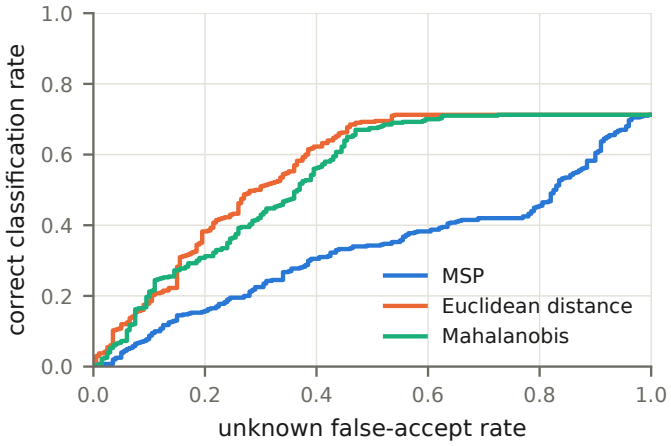


Fig. 3. OSCR curves of three scores on one network (seed 0).

E. What the MLP memorises (H9, H10)

The MLP reaches a training accuracy of 0.868 ± 0.129 but 0.251 ± 0.007 on unseen devices, the chance level of 0.25. An earlier draft of this paper read this as memorisation of the training *devices*, which a split by frame would hide. Campaign 3 tested that reading by scoring the same networks on new frames of the training devices (6720 frames per seed). H9 predicted a higher MLP accuracy there than on unseen devices. It was not supported (6/10; mean difference +0.002): the MLP scores 0.252 ± 0.006 on new frames of devices it was trained on. It memorises training frames, not devices, and fails on any new frame; the transmitter-level split is not what exposes it. Our earlier reading was wrong.

For the CNN1D the device shift is measurable but small: accuracy is 0.761 ± 0.033 on new frames of seen devices and 0.749 ± 0.032 on unseen devices (H10, 9/10; mean difference

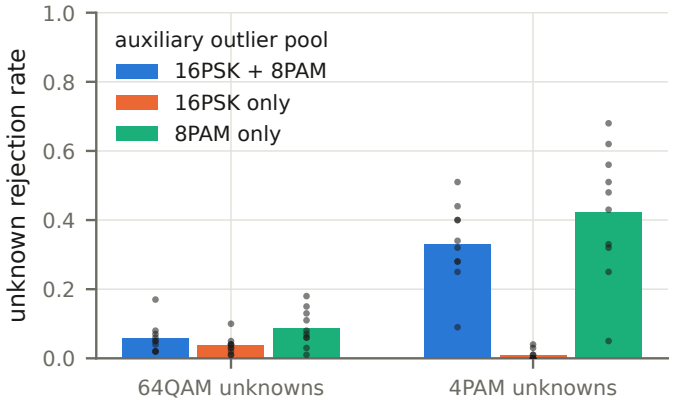


Fig. 4. Unknown rejection per unknown class for three outlier pools trained with one shared seed (campaign 3). Bars are means over ten seeds, dots are seeds; 100 frames per unknown class and seed.

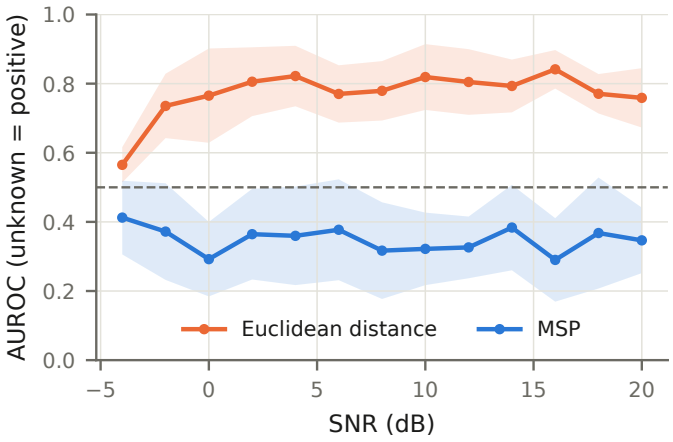


Fig. 5. AUROC per SNR for MSP and Euclidean distance on the same network; mean and standard deviation over ten seeds.

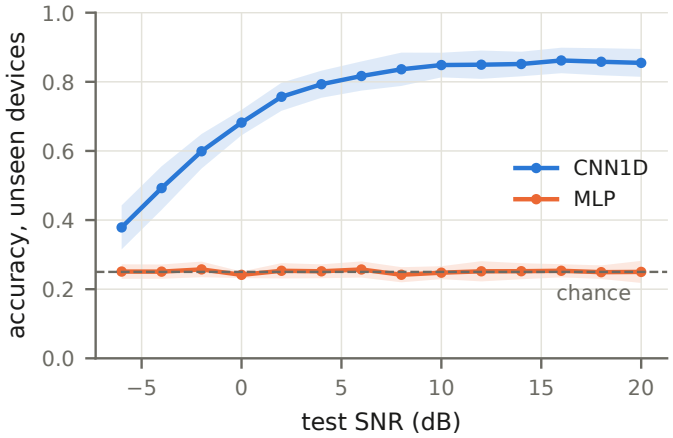


Fig. 6. Closed-set accuracy on unseen impaired devices against the test SNR; mean and standard deviation over ten seeds, 400 frames per SNR and seed.

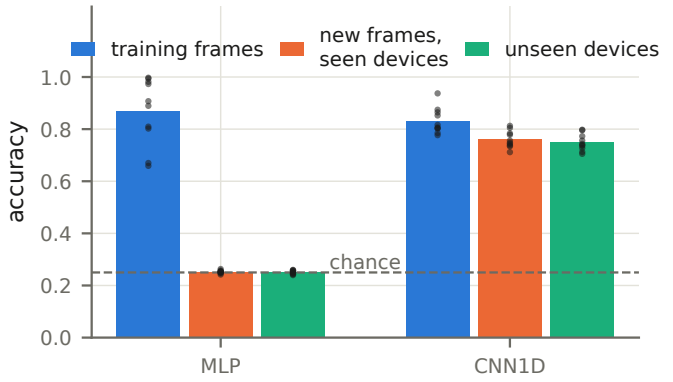


Fig. 7. Closed-set accuracy on the training frames, on new frames of the training devices and on unseen devices (campaign 3). Bars are means over ten seeds, dots are seeds.

+0.012; Fig. 7).

F. Mahalanobis shrinkage (exploratory)

Because H5 was tested at one shrinkage value, campaign 3 recomputed the Mahalanobis score on the same networks for five values of α. This analysis was chosen after seeing campaign 2 and has no pass criterion. The AUROC is 0.764, 0.748, 0.753 and 0.769 at $\alpha = 0.01$, 0.1, 0.5 and 0.9, against 0.779 for the Euclidean score, which $\alpha = 1$ reproduces exactly (Fig. 8). No tested amount of covariance whitening improved on the Euclidean score.

VII. Discussion

The pattern in Table I is that the logits are the wrong place to look for these unknowns. A 64QAM frame contains the amplitude and phase structure of 16QAM, and a 4PAM frame that of BPSK; the classifier maps both to a known class with high confidence, and every functional of the logits inherits that confidence. Changing the functional (H6) did not help. The embedding still separates the constellations, and the plain Euclidean distance uses that separation best; whitening by the within-class covariance (H5 and the sweep) removed part of it. Feature-space scores are known to outperform output-space confidence under semantic similarity [2], [9]; the benchmark quantifies this under a controlled shift and adds that the simpler distance was the better one here.

The outlier-pool experiment is a caution for outlier exposure in general. The measured gain came from the similarity between one outlier class and one unknown class, and it vanished when that outlier class was removed. A result obtained with an outlier pool therefore says little without a statement of how the pool relates to the unknowns under test.

The replication changed the conclusions in three places. H8, supported on the first five seeds with a small effect, did not survive fresh seeds. The statement that MSP is below chance "in every seed" became nine of ten. And our reading of the MLP result was refuted by the experiment designed to test it. None of these would have been visible in the first five seeds alone.

VIII. Limitations and Threats to Validity

Synthetic data. The transmitter model has no fading, multi-path, filter mismatch, timing error or amplifier memory, and no receiver model. Nothing here predicts performance on measured signals.

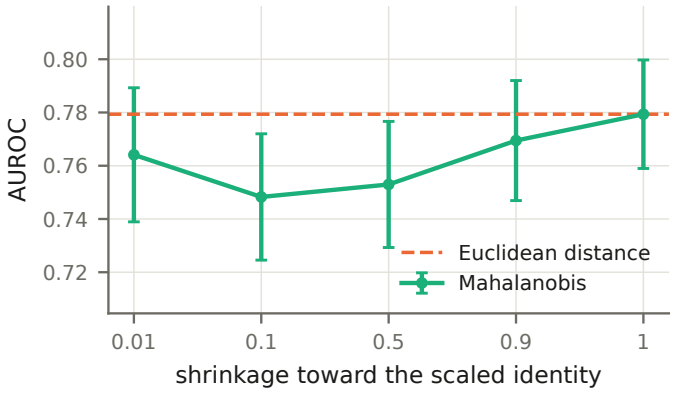


Fig. 8. Mahalanobis AUROC against the shrinkage α (exploratory); mean and standard deviation over ten seeds. $\alpha = 1$ equals the Euclidean score.

TABLE II
PRE-REGISTERED HYPOTHESES. COUNTS ARE SEEDS WITH THE PREDICTED SIGN. H7S, H9 AND H10 WERE RUN ON TEN SEEDS AT ONCE.

	Prediction	0–4	5–9	All	Outcome
H1	Euclidean > MSP (AUROC)	5/5	5/5	10/10	replicated
H2	Objectosph. > MSP (rejection)	5/5	5/5	10/10	replicated
H3	fingerprint: impaired > ideal	5/5	5/5	10/10	replicated
H4	CNN1D: ideal > impaired	3/5	4/5	7/10	inconsistent
H5	Mahalanobis > Euclidean	0/5	0/5	0/10	refuted
H6	energy > MSP	2/5	0/5	2/10	refuted
H7	4PAM rej.: 8PAM > 16PSK pool	5/5	5/5	10/10	replicated
H7s	as H7, shared training seed	–	–	10/10	supported
H8	MSP AUROC: low > high SNR	4/5	3/5	7/10	not replicated
H9	MLP: seen > unseen devices	–	–	6/10	not supported
H10	CNN1D: seen > unseen devices	–	–	9/10	supported

One architecture and one configuration. One network, one SNR grid, one calibration target and one pair of Objectosphere parameters were used. The conclusions are specific to them.

Choice of classes. The known, unknown and auxiliary classes were fixed before the first run, but by us. The results concern unknowns that are close to known constellations; unrelated unknowns may behave differently.

Statistical power. Ten seeds support only large effects. The criteria are sign counts, not significance tests, and effects of a few points (H4, H10) are at the edge of what the design resolves.

Pre-registration record. The plan of campaign 1 was committed after its run had started, as stated in Section V; its hypotheses were retested on fresh seeds under a plan committed beforehand. The shrinkage sweep is exploratory.

Correlated frames. Frames of one device are correlated, so frame counts overstate the number of independent samples; per-SNR cells are small.

IX. CONCLUSION

In a synthetic benchmark with explicit transmitter impairments and transmitter-level splits, scores computed from the logits (MSP, max-logit, energy) did not detect unknown modulations that resemble known ones: their mean AUROC was below chance on the same network on which the Euclidean distance to class centroids reached 0.779 ± 0.020. Mahalanobis scoring did not improve on the Euclidean distance at any tested shrinkage, and the benefit of training against auxiliary outliers was transfer from one outlier class to one similar unknown class. These findings held on fresh seeds. An SNR dependence of the MSP failure did not, and our interpretation of the MLP's failure was refuted by a follow-up experiment.

The benchmark is small enough to rerun in under an hour and a half on four CPU threads, and its plans, per-seed results and generators are released so that it can be extended: with a channel model, with outlier pools that are disjoint from the unknowns by construction, with further architectures, and above all with a measured dataset that has per-device labels [3], [13].

REPRODUCIBILITY

One command per campaign regenerates the per-seed results from the seeds; the numbers in this paper are generated from those results by a script, and the build fails if a number is missing. The repository also contains the three analysis plans with their commit history and an export of the data of one seed.

DISCLOSURE

An AI assistant (Claude, Anthropic) was used for software development, experiment orchestration and drafting of this manuscript.

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